MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Treatment of Malaria in Adult and Pediatric Patients

WHO recommendations are to treat uncomplicated malaria from all species with artemisinin-based combination therapy (such as Coartem™) based on safety and effectiveness of the drug, as well as to help streamline treatment recommendations for malaria. CDC and WHO treatment guidelines now concur with recommendations for Coartem™ as first line agent with known chloroquine resistance or species not identified. The CDC currently recommends chloroquine-containing regimens as first line therapy for chloroquine-susceptible species of malaria, which differs from WHO recommendations. We have elected to streamline institutional recommendations to be in line with WHO recommendations when there is discrepancy, given local drug availability and that Coartem™ clears parasitemia faster than chloroquine.

Definitions

Uncomplicated malaria:

Persons with a positive blood smear OR history of recent possible exposure and no other recognized pathology who do not meet severe criteria

Severe malaria:

Persons with a positive blood smear OR history of recent possible exposure and no other recognized pathology who have one or more of the following clinical criteria:

- Parasitemia of > 5%
- Impaired consciousness/coma
- Severe normocytic anemia
- Renal failure
- Pulmonary edema
- Acute respiratory distress syndrome
- Circulatory shock
- Disseminated intravascular coagulation
- Spontaneous bleeding
- Acidosis
- Hemoglobinuria
- Jaundice
- Repeated generalized convulsions

Chloroquine sensitivity:

- *P. falciparum* acquired in Central America (west of the Panama Canal), Haiti, and Dominican Republic
- All *P. malariae*, *P. knowlesi*, and *P. ovale*
- All *P. vivax* EXCEPT infections acquired in Papua New Guinea or Indonesia

Uncomplicated malaria	
P. falciparum or not identified	*P. malariae* or *P. knowlesi*
P. vivax or *P. ovale* chloroquine sensitive	*P. vivax* or *P. ovale* chloroquine resistant
Severe malaria	
All species	
Supply information	
Pharmacy steps to obtaining Artesunate	
Footnotes & References	

Plasmodium Spp.	Treatment Regimen	Duration	Comments
Uncomplicated malaria with *P. falciparum* or species not identified Recommend to empirically start malaria treatment for patients with Ebola PUI	<u>Chloroquine resistant, unknown, or sensitive:</u> *Preferred* **Artemether 20 mg-lumefantrine 120 mg (Coartem™)** = 1 tablet 5 - < 15 kg: 1 tablet/dose 15 - < 25 kg: 2 tablets/dose 25 - < 35 kg: 3 tablets/dose ≥ 35 kg: 4 tablets/dose The patient should receive the initial dose, followed by the second dose 8 hours later, then 1 dose PO BID for the following 2 days. *Alternative* **Atovaquone 62.5 mg-proguanil 25 mg (Malarone™)** = 1 peds tablet 5 - < 8 kg: 2 peds tablets PO daily 8 - < 10 kg: 3 peds tablets PO daily **Atovaquone 250 mg-proguanil 100 mg (Malarone™)** = 1 adult tablet 10 - < 20 kg: 1 adult tablet PO daily 20 - < 30 kg: 2 adult tablets PO daily 30 - < 40 kg: 3 adult tablets PO daily ≥ 40 kg: 4 adult tablets PO daily	**Coartem™**: 3 days **Malarone™**: 3 days	WHO recommends treatment with artemisinin-based therapies (ACTs) for all *P. falciparum* If patient used **Malarone™** as chemoprophylaxis then use another treatment option Please see CDC guidelines for treatment of pregnant women during the first trimester CDC Malaria Treatment Table **Coartem™** and **Malarone™** should be taken with food, and are OK to crush

Plasmodium Spp.	Treatment Regimen	Duration	Comments
Uncomplicated malaria with chloroquine sensitive *P. vivax* or *P. ovale*	*Preferred* **Artemether 20 mg-lumefantrine 120 mg (Coartem™)** = 1 tablet 5 - < 15 kg: 1 tablet per dose 15 - < 25 kg: 2 tablets per dose 25 - < 35 kg: 3 tablets per dose ≥ 35 kg: 4 tablets per dose The patient should receive the initial dose, followed by the second dose 8 hours later, then 1 dose PO BID for the following 2 days. *AND* **Primaquine** (after confirmation that NOT G6PD deficient AND NOT pregnant) *Adults (≥ 18 years):* 30 mg primaquine base PO daily *Children (< 18 years):* 0.5 mg/kg primaquine base PO daily (max: 30 mg/dose) *Alternative for chloroquine sensitive organisms:* **Chloroquine phosphate (Aralen™)** – dose based on chloroquine base Adults (≥ 18 years): 600 mg base PO immediately, followed by 300 mg base PO at 6, 24, and 48 hours (total dose: 1,500 mg base) Children: 10 mg/kg base (max: 600 mg) PO immediately, followed by 5 mg/kg base (max: 300 mg) PO at 6, 24, and 48 hours (total dose: 25 mg/kg base) *AND* **Primaquine** (after confirmation that NOT G6PD deficient AND NOT pregnant) *Adults (≥ 18 years):* 30 mg primaquine base PO daily *Children (< 18 years):* 0.5 mg/kg primaquine base PO daily (max: 30 mg/dose)	**Coartem™**: 3 days **Primaquine**: 14 days **Chloroquine**: 48 hours	Please see CDC guidelines for management during Pregnancy CDC Malaria Treatment Table Treatment of *P. vivax* or *P. ovale* requires addition of second agent to eliminate dormant hepatic hypnozoites There may be a role for **chloroquine** prophylaxis for patients unable to take **primaquine**, consult with ID/CDC for further guidance **Coartem™** clears parasites more quickly than chloroquine Regimens used for chloroquine-resistant species of malaria can also be substituted for chloroquine-sensitive species of malaria as needed based on drug availability. If **chloroquine** is being used, check EKG for QT interval prior to dosing, given risk for QT prolongation **Coartem™** and **Malarone™** should be taken with food, and are OK to crush **Chloroquine** comes as a liquid formation, but if this cannot be obtained, then crushing the tablet is acceptable. **Chloroquine phosphate** – 250 mg of **chloroquine phosphate** is equivalent to 150 mg of **chloroquine base** (600 mg **chloroquine base** = 1,000 mg salt; 300 mg **chloroquine base** = 500 mg salt)

Plasmodium Spp.	Treatment Regimen	Duration	Comments
Uncomplicated malaria with chloroquine resistant *P. vivax* or *P. ovale*	*Preferred* **Artemether 20 mg-lumefantrine 120 mg (Coartem™)** = 1 tablet 5 - < 15 kg: 1 tablet per dose 15 - < 25 kg: 2 tablets per dose 25 - < 35 kg: 3 tablets per dose ≥ 35 kg: 4 tablets per dose The patient should receive the initial dose, followed by the second dose 8 hours later, then 1 dose PO BID for the following 2 days. *AND* **Primaquine** (after confirmation that NOT G6PD deficient AND NOT pregnant) *Adults (≥ 18 years):* 30 mg primaquine base PO daily *Children (< 18 years):* 0.5 mg/kg primaquine base PO daily (max: 30 mg/dose) *Alternative for chloroquine-resistant organisms:* **Atovaquone 62.5 mg-proguanil 25 mg (Malarone™)** = 1 peds tablet 5 - < 8 kg: 2 peds tablets PO daily 8 - < 10 kg: 3 peds tablets PO daily **Atovaquone 250 mg-proguanil 100 mg (Malarone™)** = 1 adult tablet 10 - < 20 kg: 1 adult tablet PO daily 20 - < 30 kg: 2 adult tablets PO daily 30 - < 40 kg: 3 adult tablets PO daily ≥ 40 kg: 4 adult tablets PO daily *AND* **Primaquine** (after confirmation that NOT G6PD deficient AND NOT pregnant) *Adults (≥ 18 years):* 30 mg primaquine base PO daily *Children (< 18 years):* 0.5 mg/kg primaquine base PO daily (max: 30 mg/dose)	**Coartem™**: 3 days **Primaquine**: 14 days **Malarone™**: 3 days **Primaquine**: 14 days	Please see CDC guidelines for management during Pregnancy CDC Malaria Treatment Table Treatment of *P. vivax* or *P. ovale* requires addition of second agent to eliminate dormant hepatic hypnozoites There may be a role for **chloroquine** prophylaxis for patients unable to take **primaquine**, consult with ID/CDC for further guidance **Coartem™** clears parasites more quickly than chloroquine Regimens used for chloroquine-resistant species of malaria can also be substituted for chloroquine-sensitive species of malaria as needed based on drug availability. **Coartem™** and **Malarone™** should be taken with food, and are OK to crush

Plasmodium Spp.	Treatment Regimen	Duration	Comments
Uncomplicated malaria with *P. malariae* or *P. knowlesi*	*Preferred* **Artemether 20 mg-lumefantrine 120 mg (Coartem™)** = 1 tablet 5 - < 15 kg: 1 tablet per dose 15 - < 25 kg: 2 tablets per dose 25 - < 35 kg: 3 tablets per dose ≥ 35 kg: 4 tablets per dose The patient should receive the initial dose, followed by the second dose 8 hours later, then 1 dose PO BID for the following 2 days. *Alternative* **Chloroquine phosphate (Aralen™)** – dose based on chloroquine base Adults (≥ 18 years): 600 mg base PO immediately, followed by 300 mg base PO at 6, 24, and 48 hours (total dose: 1,500 mg base) Children: 10 mg/kg base (max: 600 mg) PO immediately, followed by 5 mg/kg base (max: 300 mg) PO at 6, 24, and 48 hours (total dose: 25 mg/kg base)	**Coartem™**: 3 days **Chloroquine**: 48 hours	These species have no known chloroquine resistance, but **Coartem™** clears parasites more quickly than chloroquine If **chloroquine** is being used, check EKG for QT interval prior to dosing, given risk for QT prolongation **Coartem™** should be taken with food, and are OK to crush **Chloroquine** comes as a liquid formation, but if this cannot be obtained, then crushing the tablet is acceptable. **Chloroquine phosphate** – 250 mg of **chloroquine phosphate** is equivalent to 150 mg of **chloroquine base** (600 mg **chloroquine base** = 1,000 mg salt; 300 mg **chloroquine base** = 500 mg salt)

Plasmodium Spp.	Treatment Regimen	Duration	Comments
Severe malaria (all spp) Severe malaria is most often caused by *P. falciparum*, but is also a risk with *P. knowlesi*	**Artesunate** (IV) is the first line recommended agent Adults and children ≥ 20 kg: 2.4 mg/kg/dose at 0, 12 hours, and 24 hours Children < 20 kg: 3 mg/kg/dose at 0, 12 hours, and 24 hours *Once parasitemia < 1%, follow IV artesunate with a full 3-day follow-on course of:* **Artemether 20 mg-lumefantrine 120 mg (Coartem™)** = 1 tablet 5 - < 15 kg: 1 tablet per dose 15 - < 25 kg: 2 tablets per dose 25 - < 35 kg: 3 tablets per dose ≥ 35 kg: 4 tablets per dose The patient should receive the initial dose, followed by the second dose 8 hours later, then 1 dose PO BID for the following 2 days. *Alternative once parasitemia < 1%, follow IV artesunate with a full 3-day follow-on course of:* **Atovaquone 250 mg-proguanil 100 mg (Malarone™)** = 1 adult tablet 10 - < 20 kg: 1 adult tablet PO daily 20 - < 30 kg: 2 adult tablets PO daily 30 - < 40 kg: 3 adult tablets PO daily ≥ 40 kg: 4 adult tablets PO daily **Atovaquone 62.5 mg-proguanil 25 mg (Malarone™)** = 1 peds tablet 5 - < 8 kg: 2 peds tablets PO daily 8 - < 10 kg: 3 peds tablets PO daily *For patients with *P. vivax* and *P. ovale* please also provide **Primaquine** (after confirmation that NOT G6PD deficient AND NOT pregnant) *Adults (≥ 18 years):* 30 mg primaquine base PO daily *Children (< 18 years):* 0.5 mg/kg primaquine base PO daily (max: 30 mg/dose)	Continue treatment with **artesunate** IV q24h until parasitemia is < 1% and able to tolerate oral medications. Maximum duration of IV **artesunate**: 7 days If there is persistent parasitemia at the end of the follow-on course, would warrant extension of **Coartem™** until negative smears x2.	If patient used **Malarone™** as chemoprophylaxis then use an another treatment option IV quinine is not available in the US. Delayed hemolysis can occur ~1 week after artesunate treatment in patients with hyperparsitemia. Recommend follow up weekly CBCs for 28 days after initiation of Artesunate. (See page 11 of IND document) Monitor carefully for hypoglycemia during the course of therapy Blood smears should be repeated every 12 to 24 hours, until at least 2 consecutive blood smears are negative. Also, blood smear should be repeated at the end of treatment. (See page 19 of IND document) **Coartem™** and **Malarone™** should be taken with food, and are OK to crush If delay in obtaining artesunate, recommend oral anti-malarial drug (**Artemether-lumefantrine**) immediately while awaiting delivery of IV Artesunate. If oral medications are not tolerated, consider administration via nasogastric tube or after an antiemetic

Footnotes:

https://www.cdc.gov/malaria/resources/pdf/Malaria_Treatment_Table_120419.pdf
CDC MALARIA HOTLINE: 770-488-7788 (M-F, 9am to 5pm), 770-488-7100 (after hours)

Last revised: 04/2021	**Antimicrobial Stewardship Subcommittee:** 03/2021	**P&T:** 04/2021

References	
	Citation
1	Centers for Disease Control and Prevention. Malaria Treatment (United States). https://www.cdc.gov/malaria/diagnosis_treatment/clinicians1.html. May 26, 2020.
2	World Health Organization. Guidelines for the treatment of malaria. 3rd edition. 2015. https://www.ncbi.nlm.nih.gov/books/NBK294440/. May 26, 2020.

Revision History	
Date	**Revision**
04/2021	Removed IND artesunate information

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.